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Enumerated Attributes — ZynqMP Reference

This is the canonical catalog of **every attribute the JTAG enumeration surfaces that matters** — what each one is, **where it lives** (BLOCK.REGISTER[.FIELD] + AXI address), its value on this board, and how to read it. It is the data dictionary behind `enumerate.tcl`, the decoders in `docs/annotations/zynqmp_{security,general}.py`, and the **Security Posture Summary** emitted by `tools/interpret.py`.

The reference board is the ZCU102 (XCZU9EG, S/N 210308BD8D4D): factory silicon with **no security provisioned**, enumerated in JTAG-idle. That makes its values the **baseline** — for the security gates (Part B) it's the all-open reference; for the platform/state attributes (Part A) it's the clean post-BootROM, pre-FSBL state. On another board the deviations from this baseline are what you read off.

This document has three parts: **Part A** platform & state attributes, **Part B** the security & trust posture (the controls that matter most), and **Part C** an annotated register hexdump — the raw captured bytes with each attribute mapped onto its address, generated from a real capture.

Conventions: - **Location** is BLOCK.REGISTER[.FIELD] · 0xADDRESS [· bits]. Bit layouts come from the Xilinx QEMU register model (`openocd/lib/zynqmp-regs-qemu.tcl`, the single source of truth) and UG1085 — no hand-invented bit fields. - **This board** = the value

observed on this unprovisioned ZCU102 in JTAG-idle.

Part A — Platform & state attributes

What the chip *is* and what state it's in. Not security gates, but they bound what every other attack/read can reach, and they're the first thing to establish on any board.

A1. Silicon identity

Attribute	Location	This board	What it tells you
Device part / family	CSU.IDCODE · 0xFFCA0040	0x24738093 (XCZU9EG)	Die + revision; the variant table (openocd/lib/zynqmp-variants.tcl) names capabilities (A53 count, GPU/VCU/RE, GEMs).
Silicon revision	CSU.VERSION · 0xFFCA0044 (PS_VERSION)	3 (production)	0/1 = ES1/ES2 engineering samples (different errata).
Per-chip unique ID	EFUSE.DNA_0..2 · 0xFFCC100C–1014	0x40000000_0170CFA7_44804345	96-bit fingerprint; may be read-gated on a hardened part.

A2. Compute elements (APU / RPU)

Attribute	Location	This board	What it tells you
APU cores	quad Cortex-A53 (per variant)	4 cores, all in reset at idle	The main application CPUs (EL0-EL3).
A53 release state	RST_FPD_APU · 0xFD1A0104 + APU.RVBARADDR0L · 0xFD5C0040	core 0 releasable → halts at 0xFFFC0000, CPSR 0x000003cd (EL3H)	Whether/where a core can be brought up over JTAG (the release primitive).
A53 EL3 system regs	per-core debug regs (when halted)	EL3-Secure, MMU off after release	Security state of a halted core (SCR_EL3, SCTLr, etc.).
RPU cores	dual Cortex-R5 (RPU.* · 0xFF9A0000)	lockstep vs split per RPU_GLBL_CNTL	Real-time cores + TCM; RPU island is PMU-gated from JTAG-idle.

A3. Clocks (PLLs and reference clocks)

Attribute	Location	This board	What it tells you
FPD PLLs	CRF_APB PLL_CTRL/STATUS (0xFD1A0020+, status 0xFD1A0044)	APLL/DPLL/VPLL up after BootROM	Full-power-domain clock sources (APU, DDR, video).
LPD PLLs	CRL_APB PLL_CTRL/STATUS (status 0xFF5E0040)	IOPLL/RPLL up	Low-power-domain clock sources.
Peripheral ref clocks	per-peripheral *_REF_CTRL (0xFF5E0050+, 0xFD1A0064/68)	most gated in idle	Which peripherals are clocked (pair with reset state for “usable now”).
APU clock	ACPU_CTRL · 0xFD1A0060	source + divisor + CLKACT	A53 operating clock.

A4. Power domains

Attribute	Location	This board	What it tells you
Power-island state	PMU_GLOBAL.PWR_STATE · 0xFFD80100	only BootROM-era domains up	Which islands (APU cores, R5, TCM, OCM banks, FP, PL, USB) are powered. PMU master control; not writable from JTAG-idle (any master).
PMU global control	PMU_GLOBAL.GLOBAL_CNTRL · 0xFFD80000	hardware-locked from software	

A5. Reset state

Attribute	Location	This board	What it tells you
FPD/APU resets	RST_FPD_APU · 0xFD1A0104	cores in reset ($\approx 0x00003d0f$)	ACPU0-3 / L2 / power-on resets — the A53 release knobs. Per-peripheral reset bits (QSPI/UART/SPI/SDIO/I2C/...). Subsystem-level resets.
LPD peripheral resets	RST_LPD_I0U2 · 0xFF5E0238	most peripherals held in reset	
LPD/DDR top resets	RST_LPD_TOP · 0xFF5E023C, RST_DDR_SS · 0xFD1A0108	held in idle	

A6. Memory map (what's reachable)

Region	Location	This board	What it tells you
OCM SRAM	0xFFFC0000 (256 KB, 4 banks)	readable/writable; FSBL/ATF land here when booted	On-chip RAM — the JTAG code-injection landing zone. Main memory; live only after FSBL brings up the controller. R5 tightly-coupled memory. PMU firmware/ROM space.
DDR	0x00000000+	uninitialized in JTAG-idle (probing it wedges AXI)	
TCM (R5)	0xFFE00000 / 0xFFE20000	gated in JTAG-idle (RPU island down)	
PMU RAM / ROM	0xFFDC0000 (LMB RAM) / 0xFFD00000 (ROM)	RAM writable from A53-EL3; ROM eFuse-locked	

A7. Debug topology (CoreSight / DAP)

Attribute	Location	This board	What it tells you
JTAG chain (TAPs)	scan chain	uscale.tap (DAP, IRLen 4) + uscale.ps (PS, IRLen 12)	Which TAPs are present (PMU TAP appears only when its eFuse policy allows). Which debug/memory access paths respond — itself a posture datum on a hardened part. The trace/debug fabric available.
Access Ports	DAP AP enumeration (discover.tcl)	AP0/AP1 = APB debug (0x44770002), AP2 = AXI mem (0x24770004)	
CoreSight components	per-AP ROM-table walk	Debug Units / CTIs / ETMs / funnels / ETB	

A8. IPI fabric

Attribute	Location	This board	What it tells you
APU IPI window	IPI APU agent at 0xFF300000 (TRIG/OBS/ISR/IMR)	idle	The APU↔PMU↔RPU↔PL mailbox channels (the APU→PMU PM-API path in booted state).

Part B — Security & trust posture

The security controls. On this dev board every one reads OFF/unprovisioned (the all-open baseline); on a hardened part they flip to show what's switched on. These are the rows the **Security Posture Summary** distills.

B1. Secure-debug exposure (most consequential)

Attribute	Location	This board	Hardened part
APU secure invasive debug	CSU.JTAG_DAP_CFG · 0xFFCA003C · bit 2 (SPIDEN)	1 (open)	0
APU secure trace	0xFFCA003C · bit 3 (SPNIDEN)	1	0
APU / RPU non-secure debug+trace	0xFFCA003C · bits 0,1,4,5	1	0 or 1
DAP / PL-TAP / PMU JTAG paths	CSU.JTAG_SEC · 0xFFCA0038 · [2:0]/[5:3]/[8:6]	0x3F (DAP+PLTAP open, PMU gated)	not 0b111

B2. Secure-boot policy

Attribute	Location	This board	Hardened part
RSA boot authentication	EFUSE.SEC_CTRL · 0xFFCC1058 · [25:11] (RSA_EN)	0	15-bit magic = enforced
Encrypt-only boot	0xFFCC1058 · bit 2 (ENC_ONLY)	0	1
Secure lockdown	0xFFCC1058 · bit 10 (SEC_LOCK)	0	1
PPK0 / PPK1 public-key hash	EFUSE.PPK0_0..11 · 0xFFCC10A0–10CC / PPK1_0..11 · 0xFFCC10D0–10FC	all-zero	non-zero (root of trust)
PPK0 / PPK1 revoked	0xFFCC1058 · [28:27] / [31:30]	0	2-bit magic = revoked
SPK ID / revocation	EFUSE.SPK_ID · 0xFFCC105C	0	partition-key revocation id

B3. Key material & confidentiality

Attribute	Location	This board	Hardened part
AES key-slot population	CSU.AES_STATUS · 0xFFCA1000 · bits 8–11 (* ZERO)	0xF00 (all empty)	a slot's bit clear = key loaded
eFuse AES key present	EFUSE.EFUSE_AES_CRC · 0xFFCC0048	0 (absent)	non-zero (key burned)
AES key read / write lock	0xFFCC1058 · bit 0 / bit 1 (AES_RDLK/WRLK)	0	1
PUF provisioned	EFUSE.PUF_CHASH · 0xFFCC1050, PUF_MISC · 0xFFCC1054	0 / 0x10000000	non-zero CHASH

B4. Anti-tamper

Attribute	Location	This board	Hardened part
Tamper response policy	CSU.CSU_TAMPER_0..12 · 0xFFCA5004–5034	all-zero (disarmed)	non-zero per armed source
Tamper events latched	CSU.TAMPER_STATUS · 0xFFCA5000	0	non-zero = fired
Software tamper trigger	CSU.CSU_TAMPER_TRIG · 0xFFCA0014	0	—

B5. Lockdown & fuse-programming policy

Attribute	Location	This board	Hardened part
eFuse write-lock	EFUSE.WR_LOCK · 0xFFCC0000	0x1 (locked until POR)	0x1
eFuse program-lock	EFUSE.EFUSE_PGM_LOCK · 0xFFCC0044	0	non-zero
eFuse ISR	EFUSE.EFUSE_ISR · 0xFFCC0030	0	programming-done/error flags
Program gates / BBRAM / error dis	0xFFCC1058 · [7:9] / bit 3 / bit 4	0	set per policy

B6. JTAG disable fuses

Attribute	Location	This board	Hardened part
JTAG disable	EFUSE.SEC_CTRL · 0xFFCC1058 · bit 5 (JTAG_DIS)	0 (JTAG on)	1 (TAPs silent)
DFT disable	0xFFCC1058 · bit 6 (DFT_DIS)	0	1

B7. Memory & peripheral TrustZone

Attribute	Location	This board	Hardened part
DDR TrustZone (ch. 0–5)	DDR_XMPU0..5.CTRL/LOCK · 0xFD000000..0xFD050000	default-region permit	regions configured + LOCK
FPD / OCM TrustZone	FPD_XMPU · 0xFD5D0000 / OCM_XMPU · 0xFFA70000	OCM CTRL 0x3	regions configured
Peripheral protection	XPPU.CTRL/ISR · 0xFF980000	enabled, no violations	apertures locked down

B8. Boot & runtime integrity

Attribute	Location	This board	Hardened part
Boot mode	BOOT_MODE_USER · 0xFF5E0200 · [3:0]	0 (JTAG idle)	SD/QSPI/eMMC/USB
Multiboot search offset	CSU.CSU_MULTI_BOOT · 0xFFCA0010	0	image-chaining offset
Auth / encrypt status	CSU.CSU_STATUS · 0xFFCA0000 (BOOT_AUTH/BOOT_ENC)	0	set after authenticated boot
BootROM / PMU ROM hash	CSU.SHA_DIGEST_0..11 · 0xFFCA2010+ / PMU_GLOBAL.ROM_VALIDATION_DIGEST_0/1 · 0xFFD80614/618	fixed per silicon revision	identifies ROM image (fingerprint)

Part C — Annotated register hexdump (where attributes reside)

Parts A/B give each attribute's location; this is the **raw view** — the captured bytes at each address with the attribute fields decoded on top, so you can see exactly where in a register dump each attribute sits. Bytes are little-endian (as a memory dump shows them); ★ marks a security control.

This is **generated from a real capture** by tools/hexdump-attributes.py (no hand-typed values). Regenerate it — and get the full dump including the platform blocks (clocks/power/reset/XMPU/XPPU/IPI/RPU) — with:

```
python3 tools/hexdump-attributes.py          # all blocks, latest capture (else golden)
python3 tools/hexdump-attributes.py --security # security blocks only
python3 tools/hexdump-attributes.py -o out.md  # write markdown to a file
```

The excerpt below is from the frozen JTAG-idle golden capture. The §4 security reads added 2026-06-08 — AES_STATUS, the CSU_TAMPER_* block, eFuse WR_LOCK/ EFUSE_ISR/PGM_LOCK/EFUSE_AES_CRC, and the full PPK0/PPK1 words — appear here once a fresh JTAG-idle capture re-freezes the baseline; re-run the tool then.

★ CSU (base 0xFFCA0000)

address	+0 +1 +2 +3	word	register	attribute fields
0xFFCA0000	00 00 00 00	0x00000000	★ CSU_STATUS	BOOT_ENC=0 BOOT_AUTH=0
0xFFCA0010	00 00 00 00	0x00000000	★ CSU_MULTI_BOOT	
0xFFCA0034	03 00 00 00	0x00000003	★ JTAG_CHAIN_STATUS	ARM_DAP=1 PL_TAP=1
0xFFCA0038	3F 00 00 00	0x0000003F	★ JTAG_SEC	SSSS_PMU_SEC=0 SSSS_PLTAP_SEC=7 SSSS_DAP_SEC=7
0xFFCA003C	FF 00 00 00	0x000000FF	★ JTAG_DAP_CFG	SSSS_RPU_NIDEN=1 SSSS_RPU_DBGEN=1 SSSS_APU_SPNIDEN=1 SSSS_APU_NIDEN=1 SSSS_APU_DBGEN=1
0xFFCA0040	93 80 73 24	0x24738093	★ IDCODE	CONST_1=1 MANUF_ID=73 PART_ID=18232 REVISION=2
0xFFCA0044	13 05 00 00	0x00000513	★ VERSION	PLATFORM=0 PS_VERSION=3

Read it as: at 0xFFCA0038 the dump holds bytes 3F 00 00 00 = 0x0000003F = JTAG_SEC, whose SSSS_DAP_SEC/SSSS_PLTAP_SEC fields are 7 (open) and SSSS_PMU_SEC is 0 (gated). At 0xFFCA003C, FF 00 00 00 = JTAG_DAP_CFG with every debug-enable bit set, including SPIDEN/SPNIDEN (secure-world debug open).

★ EFUSE (base 0xFFCC0008)

address	+0 +1 +2 +3	word	register	attribute fields
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```
0xFFCC0008 27 00 00 00 0x00000027 ★ STATUS      AES_CRC_PASS=0 AES_CRC_DONE=0 CACHE_DONE=1 EFUSE_3_TBIT=1 EFUSE_2_TBIT=1 EFUSE_0_TBIT=1
0xFFCC1058 00 00 00 00 0x00000000 ★ SEC_CTRL    RSA_EN=0 ENC_ONLY=0 JTAG_DIS=0 SEC_LOCK=0 AES_RDLK=0 AES_WRLK=0 PPK0_INVLD=0 PPK1_INVLD=0
```

SEC_CTRL reading all-zero is the whole security-boot policy at once: no RSA, no encrypt-only, JTAG enabled, nothing locked — the all-open baseline.

The complete annotated hexdump (all 106 captured registers across every block) is what `tools/hexdump-attributes.py` prints without `--security`.

Reading the Security Posture Summary

`tools/interpret.py` distills Part B into a single **Security Posture Summary** table (`rule_security_posture_summary`) — one row per control with an OFF/dev → ON/provisioned verdict. Rows shown in **bold** are the ones that most change a device's exposure when ON (RSA enforce, encrypt-only, JTAG_DIS, AES read-lock, PPK provisioned, secure debug, tamper armed). Read it first on any new board; this document is the detail behind each row, plus the Part A platform context.

PDF (shareable)

A landscape PDF of this document is at `docs/11-enumerated-attributes.pdf`. Regenerate after edits:

```
cd docs && pandoc --defaults=pandoc-attributes.yaml -o 11-enumerated-attributes.pdf 11-enumerated-attributes.md
```

(The defaults use landscape + line-wrapping so the wide attribute tables and hexdump rows fit; needs pandoc + xelatex + DejaVu fonts, same toolchain as the `whitepaper`.)

See also

- `tools/hexdump-attributes.py` — generates the Part C annotated hexdump from any capture.
- [05-enumeration-tool.md](#) — the enumerate/interpret pipeline that produces these reads.
- [whitepaper/03-enumeration-reveals.md](#) — the narrative companion (what each category enables).
- `docs/annotations/zynqmp_{security,general}.py` — the per-field/per-register decoders.
- `docs/findings/zynqmp_rules.py` — `rule_security_posture_summary` and the other rules.
- `openocd/lib/zynqmp-regs-qemu.tcl` — verified register/address/bit source of truth.